

Cambridge International AS & A Level

MATHEMATICS**9709/55**

Paper 5 Probability & Statistics 1

October/November 2025**MARK SCHEME**

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **19** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Mathematics-Specific Marking Principles

- 1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
- 2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
- 3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
- 4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
- 5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.
- 6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	More information required
	Accuracy mark awarded zero
	Accuracy mark awarded one
	Independent accuracy mark awarded zero
	Independent accuracy mark awarded one
	Independent accuracy mark awarded two
	Benefit of the doubt
	Blank Page
	Incorrect
Dep	Used to indicate DM0 or DM1

Annotation	Meaning
DM1	Dependent on the previous M1 mark(s)
	Follow through
	Indicate working that is right or wrong
Highlighter	Highlight a key point in the working
	Ignore subsequent work
	Judgement
	Judgement
	Method mark awarded zero
	Method mark awarded one
	Method mark awarded two
	Misread
	Omission or Other solution
Off-page comment	Allows comments to be entered at the bottom of the RM marking window and then displayed when the associated question item is navigated to.
On-page comment	Allows comments to be entered in speech bubbles on the candidate response.
	Judgment made by the PE
	Premature approximation
	Special case
	Indicates that work/page has been seen

Annotation	Meaning
	Error in number of significant figures
	Correct
	Transcription error
	Correct answer from incorrect working

PUBLISHED**Mark Scheme Notes**

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

- M** Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).
- B** Mark for a correct result or statement independent of method marks.
- DM or DB** When a part of a question has two or more ‘method’ steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- FT** Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.
- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
 - For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
 - The total number of marks available for each question is shown at the bottom of the Marks column.
 - Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
 - Square brackets [] around text or numbers show extra information not needed for the mark to be awarded.

Abbreviations

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no ‘follow through’ from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

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Question	Answer	Marks	Guidance
1(a)	$\left[\left(\frac{5}{6}\right)^6 \left(\frac{1}{6}\right) = 0.0558, \frac{15625}{279936}\right]$	B1	
		1	
1(b)	$\left(\frac{5}{6}\right)^2 \left(\frac{1}{6}\right) + \left(\frac{5}{6}\right)^3 \left(\frac{1}{6}\right) + \left(\frac{5}{6}\right)^4 \left(\frac{1}{6}\right)$ Or $\left(\frac{5}{6}\right)^2 - \left(\frac{5}{6}\right)^5$	M1	$(1-p)^2 p + (1-p)^3 p + (1-p)^4 p$ Or $p^2 - p^5 \quad 0 < p < 1$
	$= \frac{2275}{7776}, \quad 0.293$	A1	
		2	

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Question	Answer	Marks	Guidance
2(a)	$[4p + 0.4 = 1,] \quad p = 0.15, \quad \frac{3}{20}$	B1	May be seen used in formulae.
	$[E(X) =] \quad 1 \times 0.15 + 2 \times 0.4 + 3 \times 0.3 + 4 \times 0.15$ $[= \mathbf{0.15} + \mathbf{0.8} + \mathbf{0.9} + \mathbf{0.6} = 2.45, \quad \frac{49}{20}]$	M1	Accept un-simplified numerical expression. May be calculated in variance. $1 \times p + 2 \times 0.4 + 3 \times 2p + 4 \times p, 0 < p < 1, \sum p = 1.$
	$[Var(X) =] \quad 0.15 \times 1^2 + 0.4 \times 2^2 + 0.3 \times 3^2 + 0.15 \times 4^2 - (their\ 2.45)^2$ $[0.15 \times 1 + 0.4 \times 4 + 0.3 \times 9 + 0.15 \times 16 - 6.0025]$ $[6.85 - 6.0025]$	M1	Appropriate variance formula using <i>their</i> $(E(X))^2$ value FT their p , acceptable at the bold partially evaluated stage. Must be a numerical expression, condone $\sum p \neq 1$. Accept $0.15 + 1.6 + 2.7 + 2.4 - 2.45^2$ or 6.0025 .
	$= 0.8475, \quad \frac{339}{400}$	A1	If either or both M marks not awarded, SCB1 for correct answer WWW.
		4	

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Question	Answer	Marks	Guidance
2(b)	Method 1		
	$[P(X < 8) = 1 - (P(8, 9, 10) =]$ $1 - \{ {}^{10}C_8 (0.4)^8 (0.6)^2 + {}^{10}C_9 (0.4)^9 (0.6)^1 + {}^{10}C_{10} (0.4)^{10} \}$ $[= 1 - (0.0106168 + 0.00157286 + 0.00010486)]$	M1	One term of form ${}^{10}C_x (p)^x (1-p)^{10-x}$. With $0 < p < 1, x \neq 0$ or 10 .
		A1	Correct un-simplified expression, no terms omitted, leading to final answer. Condone omission of final bracket '}'.
	$= 0.988$	B1	$0.9875 < p \leq 0.988$.
	Method 2		
	$[P(X < 8) = P(0, 1, 2, 3, 4, 5, 6, 7) =]$ ${}^{10}C_0 (0.6)^{10} + {}^{10}C_1 (0.4)^1 (0.6)^9 + {}^{10}C_2 (0.4)^2 (0.6)^8 + {}^{10}C_3$ $(0.4)^3 (0.6)^7 + {}^{10}C_4 (0.4)^4 (0.6)^6 + {}^{10}C_5 (0.4)^5 (0.6)^5 + {}^{10}C_6$ $(0.4)^6 (0.6)^4 + {}^{10}C_7 (0.4)^7 (0.6)^3$	M1	One term of form ${}^{10}C_x (p)^x (1-p)^{10-x}$. With $0 < p < 1, x \neq 0$ or 10 .
		A1	Correct un-simplified expression, no more than 5 'middle' terms omitted, leading to final answer.
	$= 0.988$	B1	$0.9875 < p \leq 0.988$.
		3	

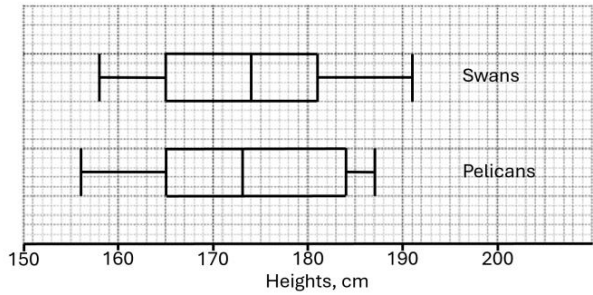
Question	Answer	Marks	Guidance
3(a)	Tree diagram with all 2, 4, 8 branches	B1	Structure, including labelling H, T, R & Bs and 4 or more correct probabilities.
	$\frac{1}{2}, \frac{1}{2}, \frac{8}{15}, \frac{7}{15}, \frac{6}{15}, \frac{9}{15}, \frac{7}{14}, \frac{7}{14}, \frac{8}{14}, \frac{6}{14}, \frac{6}{15}, \frac{9}{15}, \frac{6}{15}, \frac{9}{15}$	B1	Remaining probabilities correctly placed.
		2	
3(b)	$\frac{1}{2} \times \frac{7}{15} \times \frac{6}{14} + \frac{1}{2} \times \frac{9}{15} \times \frac{9}{15}$	M1	OE. FT P(HBB) + P(TBB) from incorrect tree diagram or $\frac{1}{2} \times \frac{a}{15} \times \frac{a-1}{14} + \frac{1}{2} \times \frac{b}{15} \times \frac{b}{15}, a=8, 7 \ b=6, 9$
	$= \frac{7}{25}, 0.28$	A1	
		2	

Question	Answer	Marks	Guidance
3(c)	$[P(1R \text{ or } 2R) = 1 - P(BB) = 1 - \frac{7}{25} = \frac{18}{25}, 0.72]$	B1	FT 1 – their 3(b) or $\frac{18}{25}$ seen.
	$P(X \cap 1R \text{ or } 2R) = \frac{1}{2} \times \frac{8}{15} \times 1 + \frac{1}{2} \times \frac{7}{15} \times \frac{8}{14} \left[= \frac{6}{15} \right]$	M1	FT from incorrect tree diagram or $\frac{1}{2} \times \frac{a}{15} \times 1 + \frac{1}{2} \times \frac{15-a}{15} \times \frac{a}{14}$ or $\frac{1}{2} \times \frac{a}{15} \times \left(\frac{a-1}{14} + \frac{15-a}{14} \right) + \frac{1}{2} \times \frac{a-1}{15} \times \frac{a}{14}$ $a = 8, 7.$
	$P(X 1R \text{ or } 2R) = \left[\frac{P(X \cap 1R \text{ or } 2R)}{P(1R \text{ or } 2R)} \right] = \frac{\frac{6}{15}}{\frac{18}{25}}$	M1	Identified values substituted into $\frac{\text{their } P(X \cap 1R \text{ or } 2R)}{\text{their } P(1R \text{ or } 2R)}$ or correct.
	$= \frac{5}{9}, 0.556$	A1	
		4	

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Question	Answer	Marks	Guidance																																																																		
4(a)	<table><thead><tr><th colspan="5">Pelicans</th><th></th><th colspan="5">Swans</th></tr></thead><tbody><tr><td></td><td></td><td></td><td></td><td>6</td><td>15</td><td>8</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td>7</td><td>5</td><td>4</td><td>0</td><td>16</td><td>2</td><td>3</td><td>5</td><td></td></tr><tr><td></td><td></td><td>8</td><td>3</td><td>1</td><td>0</td><td>17</td><td>0</td><td>0</td><td>4</td><td>4</td></tr><tr><td>7</td><td>6</td><td>5</td><td>4</td><td>2</td><td>2</td><td>18</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td>19</td><td>1</td><td></td><td></td><td></td></tr></tbody></table> Key: 1 17 8 represents 171 cm for Pelicans and 178 cm for Swans	Pelicans						Swans									6	15	8							7	5	4	0	16	2	3	5				8	3	1	0	17	0	0	4	4	7	6	5	4	2	2	18	0	1	2	3							19	1				B1	Correct stem, ignore extra values (not in reverse, not split) If a split stem-and-leaf plot is used (i.e. stem values are repeated) the remaining B marks are available.
	Pelicans						Swans																																																														
					6	15	8																																																														
			7	5	4	0	16	2	3	5																																																											
			8	3	1	0	17	0	0	4	4																																																										
7	6	5	4	2	2	18	0	1	2	3																																																											
						19	1																																																														
B1	Correct Pelicans labelled on left, leaves in order from right to left and lined up vertically no commas or other punctuation.																																																																				
B1	Correct Swans labelled on same diagram, leaves in order from right to left and lined up vertically, no commas or other punctuation. Penalise each error only once in question. E.g. commas in both sets of data. Missing or incorrect leaves can be penalised in both cases.																																																																				
B1	Correct key, for their diagram, need both team names and ‘cm’ at least once here, or in leaf headings or a suitable title. <u>If 2 separate diagram drawn</u> max marks: B1 if both stems correct, B1 if Pelicans correct to the left of the stem, B1 if both keys correct including ‘cm’ and team name.																																																																				
4																																																																					
4(b)	Median = 173 (cm)	B1	Accept Q2, must be identified.																																																																		
	[IQR =] 184 – 165	M1	$182 \leq UQ \leq 185$ – $164 \leq LQ \leq 167$ Implied if both quartile values are stated and appropriate IQR calculated accurately.																																																																		
	= 19 [cm]	A1	WWW. If M0 scored, SC1 for 19 WWW.																																																																		
		3																																																																			

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Question	Answer	Marks	Guidance												
4(c)	Box-and-whisker plot on provided grid <table border="1"><tr><td>Swans</td><td>158</td><td>165</td><td>174</td><td>181</td><td>191</td></tr><tr><td>Pelican</td><td>156</td><td>165</td><td>173</td><td>184</td><td>187</td></tr></table> Condone freehand lines Check accuracy in the middle of the vertical line using daylight rule 	Swans	158	165	174	181	191	Pelican	156	165	173	184	187	B1	All 5 key values for Swans plotted accurately in standard format using a linear scale with 3 identified values. Labelled Swans. Vertical lines may be omitted at both limits.
	Swans	158	165	174	181	191									
	Pelican	156	165	173	184	187									
	B1FT	All 5 key values for Pelicans, FT from 4b or correct , plotted accurately in standard format using a linear scale with 3 identified values. Labelled Pelicans. Vertical lines may be omitted at both limits													
	B1	Whiskers not through box for both , not drawn at corners of boxes, single linear scale for the diagram and labelled “heights” (OE) and ‘cm’ .													
	- If only one plot is labelled, assume other plot is as anticipated and mark as in scheme. - If neither of 2 plots is labelled, SCB1 for 1 fully ‘correct’ set of values plotted. - If no scale or incorrect linear scale stated, SCB1 if both correct relative to each other.														
3															
4(d)	Swans are taller or Swans’ heights more consistent. If they say Pelicans’ heights are more consistent, they need to refer to the Range	B1	Appropriate comment about heights.												
		1													

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Question	Answer	Marks	Guidance
5(a)	6S 8C_6 (= 28)	M1	One product using 2 or 3 combinations ${}^8C_x \times {}^5C_y \times {}^{12}C_z$ with $x + y + z = 6$ (one of y or z may be 0) and linked to a correct identified scenario.
	5S 1H ${}^8C_5 \times {}^5C_1$ (= 280)		
	5S 1T ${}^8C_5 \times {}^{12}C_1$ (= 672)	B1	2 correct identified outcomes, accept un-simplified. Identification can be implied by un-simplified expressions.
	4S 2H ${}^8C_4 \times {}^5C_2$ (= 700)		
	4S 1H 1T ${}^8C_4 \times {}^5C_1 \times {}^{12}C_1$ (= 4200)	M1	Add values of 6 correct scenarios, no incorrect/repeated scenarios, no extra scenarios. Identification can be implied by un-simplified expressions.
	3S 2H 1T ${}^8C_3 \times {}^5C_2 \times {}^{12}C_1$ (= 6720)		
	2 nd and 3 rd can be combined as ${}^8C_5 \times {}^{17}C_1$ and this counts as 2 scenarios for the second M1		
	Total = 12600	A1	If either or both Ms not awarded, SCB1 for 12600 WWW.
		4	
5(b)	Method 1 Total arrangements – arrangements with 3 hurdlers together		
	[With 3 hurdlers not together] $8! - (6! \times 3!)$	B1	$6! \times 3!$ seen as a term added or subtracted.
		M1	$8! - (6! \times k)$, $k \geq 1$, k is an integer.
	= 36000	A1	
	Method 2 Summing no of ways with 2 hurdlers together or all separate		
	$6C3 \times 3! \times 5! + 6C2 \times 2 \times 3! \times 5!$ [= 14400 + 21600]	B1	$6Cx \times 5!$ or $6Px \times 5!$ seen, $x = 2$ or 3 seen.
		M1	$(6C3 \times 5! + 6C2 \times 2 \times 5!) \times k$, or $6P3 \times 5! + 6P2 \times 5! \times k$ $k \geq 1$, k is an integer.
	= 36000	A1	
		3	

Question	Answer	Marks	Guidance
5(c)	Method 1		
	$5! \times 3! \times 4$	B1	5! seen in a single term method or as a common factor in a multi term method, no dividing.
		M1	3! seen in a single term method or as a common factor in a multi term method, no dividing.
	2880	A1	
		3	

Question	Answer	Marks	Guidance
6(a)	$[P(X > 55.1)] = P\left(Z > \frac{55.1 - 46.3}{6.4}\right) [= P(Z > 1.375)]$	M1	Use of $\pm$ standardisation formula with 55.1, 46.3 and 6.4 substituted appropriately, no continuity correction, allow $\sigma^2(6.4^2)$ or $\sqrt{\sigma}(\sqrt{6.4})$.
	$[= 1 - \Phi(1.375)]$ $= 1 - 0.9155$	M1	Calculating the appropriate probability area (leading to the final answer, expect <0.5).
	$= 0.0845$	A1	$0.0845 \leq z \leq 0.0846$ If one or both M marks not awarded, SCB1 for $0.0845 \leq z \leq 0.0846$.
		3	

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Question	Answer	Marks	Guidance
6(b)	$[P(X > 4.86) = P\left(Z > \frac{48.6 - 39.8}{\sigma}\right) = 0.10,$ $P\left(Z < \frac{48.6 - 39.8}{\sigma}\right) = 0.90]$	B1	± 1.282 seen, CAO, critical value.
	$\frac{48.6 - 39.8}{\sigma} = 1.282$	M1	$\pm$ standardisation formula with 48.6, 39.8 and σ equating to a z-value (not 0.10, 0.90, 0.5398, 0.4602, 0.8159, 0.1841, $1 - \text{their z-value}$), not σ^2 , condone continuity correction of ± 0.05 .
	$\left[\sigma = \frac{8.8}{1.282}\right] 6.86$	A1	AWRT. Not dependent on B mark being awarded. Condone 6.87.
		3	

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Question	Answer	Marks	Guidance
6(c)	$[\text{Mean} = 0.1 \times 150 =] 15$ $[\text{Variance} = 150 \times 0.1 \times 0.9 =] 13.5$	B1	15 and 13.5 seen, allow un-simplified. May be seen in standardisation formula. $(\sigma = \sqrt{13.5}, \frac{3\sqrt{6}}{2} \quad 3.674 \leq \sigma < 3.675 \text{ implies correct variance.}$ Withhold mark if variance clearly identified as standard deviation, condone $N(15, \sqrt{13.5})$ if standardisation formula correct or variance/standard deviation correctly stated as well.
	$[P(X < 20) =] P\left(Z < \frac{19.5 - 15}{\sqrt{13.5}}\right) [= P(Z < 1.2247)]$	M1	Using 20 (with or without a cc) and their mean and sd found using 150 in $\pm$ standardisation formula, not σ^2 .
		M1	Using continuity correction 19.5 or 20.5 in a standardisation formula. Note: $\frac{\pm 4.5}{\sqrt{13.5}}$ or $\frac{\pm 4.5}{3.674}$ gains M2 .
	$[\Phi(1.2247)] = 0.8897$	M1	Appropriate area Φ , from final process. Must be a probability. Note: correct final answer implies this M1 .
	$= 0.890$	A1	AWRT.
		5	